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Expert Systems for Structural Design

by

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用于结构设计的专家系统

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Expert Systems For Structural Design

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1. Abstract:

This paper considers the role the computer can have is less formalized phases of the structural design process, particularly during preliminary design. Since preliminary design is based largely on the designers experience, an expert systems approach to developing computer aids is appropriate. Four expert systems are presented: HI-RISE, FLODER, LOCATOR, and STRUPLE.

2. Introduction

Structural design is a process that encompasses all activities between the defintion of a need to resist bads through the construction of the resulting structure. The most common reference to *structural design* includes the activities between the defintion of a structural configuration and a detailed specification of the structure to be constructed. These activities include the transformation of a representation of a physical structure to a mathematical model, the analysis of the model, and the interpretation of the results followed by modifications to the physical representation and/or the mathematical model. These activities are commonly referred to as the design/analysis cycle.

The computer plays an important role in the stnjctural design process by extending the computational facilities of the structural designer so that he may precisely consider complex structural systems. Traditionally, the role of the computer has been restricted to the design/analysis cycle. Software packages have been developed to facilitate the definition of a model of a structural system and the analysis of the. model. This paper considers the role the computer may play in other, less formalized, phases of the structural design process, particularly in the preliminary decision making part of the process.

The paper is organized into two major parts. The first part discusses the nature of preliminary structural design. The second part presents four expert systems, developed at Carnegie Mellon University, that explore and define the decisions made during preliminary design. The expert systems presented are HI-RISE, FLODER, LOCATOR, and STRUPLE. The paper closes with some remarks on the potential for developing computer aids for preliminary structural design.

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用于结构设计的专家系统

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1. 摘要：

本文探讨了计算机在结构设计过程的非形式化阶段，尤其是初步设计阶段中所能发挥的作用。鉴于初步设计主要依赖于设计师的经验，采用专家系统方法来开发计算机辅助工具是恰当的。文中介绍了四个专家系统：HI-RISE、FLODER、LOCATOR 和 STRUPLE。

2. 引言

结构设计是一个涵盖从明确抗力需求到最终建成相应结构的全过程的活动。在最常见的语境中，结构设计特指从确定结构方案到给出待建结构的详细构造说明之间的各项活动。这些活动包括将物理结构的表达形式转化为数学模型、对模型进行分析、解读分析结果，并据此对物理表达或数学模型作出修改。上述过程通常被称为设计—分析循环。

计算机在结构设计过程中发挥着重要作用，它通过扩展结构设计师的计算能力，使其能够精确地分析复杂的结构体系。传统上，计算机的作用主要局限于设计与分析这一循环环节。为此，已开发出多种软件工具，以方便用户建立结构系统的模型并对其进行分析。本文则探讨了计算机在结构设计过程中的其他、较为非标准化的阶段所可能发挥的作用，尤其是在设计过程的初步决策阶段。

本文共分为两个主要部分。第一部分探讨了初步结构设计的本质；第二部分介绍了四套由卡内基梅隆大学开发的专家系统，这些系统用于研究并明确在初步设计阶段所作出的各项决策。所介绍的专家系统包括HI-RISE、FLODER、LOCATOR和STRUPLE。最后，本文就开发用于初步结构设计的计算机辅助工具的潜在可能性进行了简要评述。

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3. Preliminary Structural Design

The structural design process starts with the definition of a need to transmit loads in space to a support or foundation, subject to constraints on cost, geometry, and other criteria. The final product of the design process is the detailed specification of a structural configuration capable of transmitting these loads with the appropriate levels of safety and serviceability. The design process may be viewed as a sequence of three stages, as described befow.

1. PRELIMINARY DESIGN (conceptual design) involves the synthesis of potential structural systems satisfying a few key constraints, and the selection of one, or at most a few, systems to be pursued further. Synthesis requires a knowledge of structural subsystems and their appropriateness for different situations.
2. ANALYSES is the process of modeling the selected structural system and determining its response to external effects. This process involves transforming a physical structure to a mathematical model, analyzing the model, and interpreting the results of the analysis in terms of the actual physical structure.
3. DETAILED DESIGN is the selection and proportioning of the structural components such that all applicable constraints are satisfied.

During preliminary design, the form of the design solution is identified. During analysis and detailed design this form is refined. There may be significant deviations between the properties of components assumed at the analysis stage and those determined at the detailed design stage, which would necessitate a reanalysis. Other major and minor cycles of redesign may also occur. The process continues until a satisfactory (or optimal) design is obtained. The *conceptualize-analyze-detail* cycle is typical of many design paradigms.

Many of the conceptual aspects of structural design are embodied in the preliminary design phase. At this point, the only information available to the designer are the requirements of the end product. It is during this part of the design process that the creativity and experience of an engineer are needed. The increasing complexity of structural systems has magnified the importance of the preliminary design solution while, at the same time, made the synthesis of feasible solutions more difficult. Preliminary structural design can become critical to the total design process unless approached in a structured and organized fashion.

In order to find the most suitable structural system under the given conditions designers should be able to think in an overall way, concentrating on the relationship between spatial forms and structural systems and ignoring trivial details. On the other hand, designers should also be able to distinguish the details which must be considered before the properties of the whole system can be well understood. This requires that good designers be knowledgeable about architecture and structures, and also have experiences that enable them to make correct judgments and trade-offs as early as possible to avoid major conflicts later in the design process.

3. 初步结构设计

结构设计过程始于对在空间中将荷载传递至支承或基础这一需求的界定，并需满足造价、几何形态及其他相关约束条件。设计过程的最终成果是对一种能够以适当的可靠性和使用性能传递这些荷载的结构构型所作的详细规定。该设计过程可被视为由三个阶段组成的序列，具体如下所述。

1. 初步设计（概念设计）涉及对满足若干关键约束的潜在结构体系进行综合，并从中选择一个或最多几个体系以进一步开展研究。综合过程需要掌握各子结构系统的相关知识及其在不同工况下的适用性。 2. 分析是指对所选定的结构体系进行建模，并确定其对外部作用的响应。该过程包括将实际结构转化为数学模型、对模型进行分析，以及将分析结果与实际结构联系起来进行解释。 3. 详细设计是指对结构构件进行选型与配比，以确保所有适用的约束条件均得到满足。

在初步设计阶段，确定设计方案的基本形式；在分析与详细设计阶段，对该形式进行细化。在分析阶段所假设的各部件性能与在详细设计阶段最终确定的性能之间可能存在显著偏差，这将导致需要重新进行分析。此外，还可能经历其他多次大小不一的重新设计循环。这一过程持续进行，直至获得令人满意的（或最优的）设计方案。概念化—分析—细化循环是许多设计范式中的典型流程。

结构设计的许多概念性要素均体现在初步设计阶段。在这一阶段，设计者所掌握的唯一信息仅为最终产品的功能需求。正是在设计过程的这一环节，工程师的创造力与经验显得尤为重要。随着结构体系复杂性的日益增加，初步设计方案的重要性愈发凸显，与此同时，可行方案的综合与优化也变得更加困难。若不以结构化、系统化的方式开展工作，初步结构设计便可能对整个设计流程产生决定性影响。

为了在给定条件下确定最适宜的结构体系，设计师应当具备全局性思维，重点关注空间形态与结构体系之间的关系，同时忽略琐碎的细节。另一方面，设计师还应能够识别那些必须予以考虑的关键细节，只有充分理解这些细节，才能准确把握整个系统的特性。这要求优秀的设计师既精通建筑学与结构学知识，又具备丰富的实践经验，从而能够在设计初期做出正确的判断与权衡，以避免在后续设计阶段出现重大冲突。

In fact, many major issues *are* settled before preliminary structural design at the schematic level when designers, usually architects working in collaboration with engineers of related fields, generate the spatial forms of buildings based on the considerations of the building function, potential structural systems, and all other design constraints. Unfortunately, as technology advances and complexity of modern building designs increases, architects, structural engineers, and other engineers of different fields become more and more specialized, confining themselves in their own areas. This can cause communication problems among the different designers of buildings at the early stage of the building design, which in turn results in inefficient use of knowledge, time and energy, and potential major conflicts in the following design processes.

Preliminary design is achieved by synthesizing potential alternative designs. There is no standard approach to the synthesis process suitable for all design problems. One approach is to decompose the design problem into the design of independent subsystems. The nature of these subsystems will depend on the nature of the problem at hand. In a similar manner, each subsystem is further decomposed into major components. Alternative design solutions can be synthesized by considering possible combinations of the various subsystems that result from combinations of lower level components. This hierarchical approach to the synthesis of a solution enables the designer to consider a large set of possibilities based on the manner in which the subsystems and the lower level components are defined. These definitions will depend on the nature of the particular problem as well as the engineer performing the design. A particular problem may justify the decomposition of a problem from an abstract level down to a set of detailed subsystems. Another design may be better approached by considering the details first and building up to more general systems. A correct selection of this sequence can increase the efficiency of the synthesis process.

A key consideration in the synthesis of design alternatives is the identification and satisfaction of constraints at the various levels of abstraction. These constraints control the qualification of various components of the design as well as the feasibility of combinations of such components. Typically, these constraints are based on the experience of the designer, but may also include client's specifications and regional restrictions. The formal identification of these constraints is a difficult process that is never completed because as more design experience is acquired, the set of constraints may need to be modified and expanded.

Although designers may not use an exhaustive, organized approach to the synthesis of alternative preliminary designs, they do enumerate the alternatives that appear to be the best of all. This ability to identify feasible designs with partial information has not been formalized; therefore, it is difficult to pass

事实上，许多重大问题在初步结构设计的方案阶段便已得到解决。此时，设计师——通常是建筑师与相关专业工程师协同工作——基于建筑功能、潜在的结构体系以及各项设计约束条件，生成建筑的空间形态。然而，随着技术的进步和现代建筑设计复杂性的日益增加，建筑师、结构工程师及其他各专业工程师的专业化程度不断提高，逐渐局限于各自的专业领域。这可能导致在建筑设计的早期阶段，不同专业的设计师之间出现沟通障碍，进而造成知识、时间和精力的低效利用，并在后续的设计过程中引发潜在的重大冲突。

初步设计是通过综合多种潜在的备选设计方案来实现的。对于所有设计问题，并不存在一种适用于所有情形的标准综合方法。一种常用的方法是将设计问题分解为若干相互独立的子系统设计，而这些子系统的具体构成则取决于所面临问题的性质。类似地，每个子系统可进一步分解为若干主要部件。通过考虑由低层级部件组合而成的各子系统之间的可能组合，即可综合出多种备选设计方案。这种自上而下的分层式方案综合方法使设计者能够基于子系统及低层级部件的具体定义，系统性地评估大量可能性。这些定义既取决于具体问题的性质，也与执行设计的工程师密切相关。对于某些问题，从抽象层面逐步细化至一系列详细子系统的设计分解可能是合理的；而对于另一些问题，则可能更适合先关注细节，再逐步构建更为通用的系统。合理选择这一分解与综合的顺序，有助于提高设计综合过程的效率。

在设计备选方案的综合过程中，一个关键考量是在不同抽象层次上识别并满足各类约束条件。这些约束条件不仅决定了设计中各组件的合格性，也影响着这些组件组合的可行性。通常，这些约束条件主要基于设计者的经验，但也可能包括客户的技术规格和地区性限制。对这些约束条件进行形式化的识别是一项复杂且持续的过程，因为随着设计经验的积累，约束条件的集合可能需要不断调整和扩展。

尽管设计人员在生成备选初步设计方案时未必采用详尽且系统化的方法，但他们确实会枚举那些看似最优的备选方案。这种基于部分信息识别可行设计方案的能力尚未被 formalized；因此，难以进行 transfer。

this knowledge to new engineers or to develop intelligent computer assistants to this stage of the design. The use of expert system techniques holds promise for the formalization of preliminary structural design.

4. Expert System Applications to Structural Design

This section describes four expert system applications to structural design developed at Carnegie Mellon University. All four systems address the preliminary stage of design during which decisions on structural system type, material and location are made. The first three systems, HI-RISE, FLODER, and LOCATOR address the synthesis process using a generate and test paradigm. The fourth system, STRUPLE, addresses the pre-synthesis stage during which the most promising structural system types and materials are identified using analogical reasoning.

Although these applications are referred to as expert systems, the developers did not formally interview experts prior to implementation. These expert systems have many of the characteristics of current expert system technology, including separation of control and domain knowledge, use of symbollic reasoning, and certain heuristics gathered from publications and local designers.

4.1. HI-RISE

HI-RISE [5] is a knowledge based expert system that performs the preliminary structural design of high rise buildings. Given the space planning of a building, which is described by a three dimensional grid, HI-RISE generates and presents alternative feasible structural systems. HI-RISE is implemented in PSRL [7], a combination of a production system and frame representation language. The declarative knowledge is represented in the form of frames and rules and the procedural knowledge in the form of Lisp functions.

The motivation for developing a system such as HI-RISE centers around the need for the engineer to identify and evaluate more design alternatives during the configuration stage. Many times the most obvious solution is selected because the engineer does not have the time or the money to develop multiple configurations. A system such as HI-RISE could take advantage of the speed and memory of a computer to quickly generate multiple, feasible configurations. The intention of HI-RISE is to provide an intelligent design assistant to suggest alternatives, not to automate the preliminary design process.

The input to HI-RISE is a three dimensional grid, specifying potential locations for beams, columns, and walls. The grid represents the result of space planning and the architectural restrictions. The topology of the grid is defined by the number of stories and the number of bays in each direction. The geometry is defined by the dimensions of the bays and the minimum required clearance for a typical story. Other spatial constraints, such as the location of service shafts or internal spaces, are specified in terms of their

将这些知识传授给新入职的工程师，或开发用于设计这一阶段的智能计算机辅助系统。专家系统技术的应用为初步结构设计的规范化提供了广阔前景。

4. 专家系统在结构设计中的应用

本节介绍了卡内基梅隆大学开发的四个用于结构设计的专家系统应用。这四个系统均针对设计的初步阶段，该阶段主要确定结构体系类型、材料及布置方案。前三个系统——HI-RISE、FLODER 和 LOCATOR——采用“生成与检验”的范式来解决结构方案的综合问题。第四个系统 STRUPLE 则聚焦于预综合阶段，通过类比推理识别最具潜力的结构体系类型和材料。

尽管这些应用被称为专家系统，但开发人员在实施之前并未正式访谈专家。这些专家系统具备当前专家系统技术的诸多特征，包括控制知识与领域知识的分离、符号推理的运用，以及从文献和本地设计人员处获取的若干启发式规则。

4.1 高层建筑

HI-RISE [5] 是一个基于知识的专家系统，用于进行高层建筑的初步结构设计。给定以三维网格描述的建筑空间布局后，HI-RISE 会生成并呈现若干可行的结构方案。HI-RISE 基于 PSRL [7] 实现，该系统融合了产生式系统与框架表示语言。其中，陈述性知识以框架和规则的形式表示，而过程性知识则以 Lisp 函数的形式表达。

开发HI-RISE这类系统的主要动机在于，工程师在配置阶段需要识别并评估更多的设计方案。很多时候，最显而易见的方案会被直接采纳，因为工程师既没有足够的时间，也没有足够的预算来制定多种配置方案。像HI-RISE这样的系统可以充分利用计算机的运算速度和存储能力，快速生成多个可行的配置方案。HI-RISE的初衷是提供一个智能设计助手以建议备选方案，而非实现初步设计过程的自动化。

HI-RISE 的输入是一个三维网格，用于指定梁、柱和墙体的潜在布置位置。该网格反映了空间规划及建筑约束的结果。网格的拓扑结构由楼层数以及各方向上的开间数决定；其几何尺寸则由各开间的尺度及典型层所需的最小净空高度确定。其他空间约束条件，例如设备竖井或内部空间的位置，则以其

location on the input grid. Other input information required by HI-RISE is the intended occupancy of the building, and the wind and live load.

HI-RISE is currently restricted to buildings with a rectangular perimeter and a simple rectangular structural grid. The knowledge incorporated in HI-RISE is appropriate for buildings between 5 and 50 stories. These restrictions were imposed to allow a study of the feasibility of developing a computer aid for preliminary design. Many of these restrictions have been relaxed in more recent attempts to develop intelligent computer aids to preliminary structural design.

In HI-RISE, the preliminary design process is divided into two major tasks; each task addresses the design of a functional system. The functional systems are designed in a fixed order: first the lateral load resisting system, followed by the gravity load resisting system. Each of the two major tasks are deomposed into a set of similar subtasks. The subtasks have the same goals for each functional system, however, the details of reaching these goals differ. The subtasks are described below.

- *Synthesis.* The first task is to synthesize alternatives for the functional system under consideration. The synthesis is performed as a depth first search through the appropriate generic subsystems stored in the knowledge base. Each subsystem has a set of synthesis levels with itemized alternatives at each level. For example, the levels and alternatives for the lateral load resisting systems are: 3D-lateral, core or 2D orthogonal; 2D-lateral, rigid frame, braced frame, or shear wall; material, steel or reinforced concrete. Synthesis is implemented as sets of rules that represent the decisions to select an alternative at a given level, move to a lower level, move to a higher level, and eliminate an infeasible design alternative.
- *Analysis.* The purpose of the analysis subtask is to determine the feasibility of an alternative and to define its component groups. Feasibility is determined by the formulation and evaluation of one or more feasibility constraints. An approximate analysis provides one set of ingredients, namely, the required load capacity of the system components. Component groups are defined so that preliminary sizing or proportioning is performed only for one component in a group.
- *Evaluation.* Evaluation of a structural design may be based on many diverse features of the design. Evaluation is usually done by designers in an abstract form and is usually based on cost. Some of the other features considered are esthetics, efficiency, and structural integrity. HI-RISE evaluates alternatives with a linear evaluation function. There is a distinct function for each of the functional systems. The variables in the function are features of the system that may be quantified, e.g. drift.
- *System Selection.* HI-RISE presents all structurally feasible systems to the user, indicating which system has been determined to be the $^{M}best^{M}$, selected as the system with the minimum value assigned by the evaluation function. The user may either accept the recommended design or override the decision of HS-RISE and choose one of the other structurally feasible systems.

Structural systems are represented as hierarchically defined frames that are used as templates for generating alternative configurations. A frame is an object with declarative and procedural attributes. The frames can be considered in three catergories: global, functional, and physical. The global frames, shown below, are the *building* and *grid* frames. These frames store information global to the entire building. The

输入网格上的位置。Hl-RISE所需的其他输入信息包括建筑的预期用途，以及风荷载和活荷载。

HI-RISE 目前仅适用于具有矩形平面轮廓和简单矩形结构网格的建筑。HI-RISE 中所集成的知识适用于 5 至 50 层的建筑。这些限制的设定旨在便于开展针对初步设计计算机辅助工具可行性研究。在近年来开发用于结构初步设计的智能计算机辅助工具的过程中，其中许多限制已得到放宽。

在HI-RISE中，初步设计过程被划分为两项主要任务，每项任务分别针对一个功能系统的设计。各功能系统的设计按照既定顺序进行：首先设计抗侧力体系，随后设计抗重力体系。这两项主要任务均被分解为一组相似的子任务。对于每个功能系统，这些子任务具有相同的目标，但实现这些目标的具体细节有所不同。下文将对这些子任务进行详细说明。

- 综合。首要任务是为所考虑的功能系统生成备选方案。该综合过程以深度优先搜索的方式，遍历知识库中存储的相应通用子系统。每个子系统均具有一组综合层级，且每一层级均列有具体的备选方案。例如，抗侧力体系的层级及备选方案包括：三维抗侧力体系——核心筒或二维正交体系；二维抗侧力体系——刚架、支撑框架或剪力墙；材料——钢结构或钢筋混凝土。综合过程通过一组规则来实现，这些规则用于表征在给定层级上选择某一备选方案、向下一层级推进、向上一层级回溯以及剔除不可行设计方案等决策。

- 分析。分析子任务的目的是确定某一替代方案的可行性，并明确其组成部分的分组。可行性的判定基于一个或多个可行性约束条件的制定与评估。近似分析提供了一组基本要素，即系统各组件所需承受的载荷能力。组件分组的设定旨在使初步的尺寸确定或配比仅针对每个分组中的一个组件进行。

- 评估。结构设计的评估可基于设计的诸多不同特征进行。评估通常由设计人员以抽象形式开展，且主要依据成本。此外，还需考虑其他一些特征，如美观性、效率以及结构完整性。HI-RISE 采用线性评估函数对备选方案进行评价，其中每个功能系统均对应一个独立的评估函数。该函数中的变量为可量化的系统特征，例如侧向位移。

- 系统选择。Hl-RISE 向用户呈现所有在结构上可行的系统，并指出已被确定为最优的系统，即由评价函数赋予最低值的系统。用户可以选择接受推荐的设计，也可以覆盖 Hl-RISE 的决策，从其他结构上可行的系统中自行选择一个。

结构系统被表示为分层定义的框架，这些框架用作生成备选配置的模板。框架是一种同时具有声明性属性和过程性属性的对象。框架可分为三大类：全局框架、功能框架和物理框架。下文所示的全局框架包括建筑框架和网格框架，这些框架存储与整个建筑物相关的全局信息。

building frame stores information about the occupancy and the design loads. The grid frame stores information about the structural grid, such as the number of stories, the number of bays in each direction, and their dimensions. There are procedural attachments to some of the grid attributes to automatically store information about ratios and total dimensions when the appropriate informatbn is provided to infer these values.

```
{  building
            occupancy
            wind-load
            live-load }

{  grid
            part-of building
            stories
            story-dim
            min-clear
            narrow-bays
            narrow-dim
            wide-bays
            wide-dim
            mech-floor
            shaft
            shaft-sym ... }
```

The functional frames are the *lateral* and *gravity* frames, serving to divide the structural systems according to function. Only those structural systems responsible for resisting lateral loads are linked to the lateral frame, and gravity systems to the gravity frame. These frames contain information about function and status of alternative configurations.

```
{  lateral
            part-of grid
            best-lat  ... }

{  gravity
            part-of  grid
            uses   lateral
            best-grav   ... }
```

The frames that represent the physical level are hierarchically defined according to function. The *3D-lateral*$_{\%}$ *2D-lateral,* and *material* frames represent decisions made for the configuration of the lateral load resisting system. The *20-horizontal, support-edges,* and *support-div* frames represent decisions made for the configuration of the gravity, or floor, system. There are additional frames in the physical level that represent the information associated with components such as beams, columns, and diagonals.

```
{  3D-lateral
            is-alt  lateral
            3D-descrip'tion }
```

建筑框架存储有关建筑物使用情况及设计荷载的信息。网格框架则存储与结构网格相关的信息，例如楼层数、各方向的开间数及其尺寸。针对部分网格属性，还设置了过程性关联，当提供相应信息以推导这些值时，可自动存储有关比值及总尺寸的信息。

建筑物人员荷载、风荷载、活荷载

{ 网格

建筑楼层的一部分，楼层尺寸最小净空

窄跨 窄深 宽跨 宽深 机械层 井道 井道对称 ...

功能框架包括抗侧力框架和重力框架，用于按功能对结构体系进行划分。只有承担抗侧力作用的结构体系才与抗侧力框架相连，而重力体系则与重力框架相连。这些框架包含了关于备选方案配置的功能与状态信息。

{ 格网的横向部分采用最佳横向参数 ... } { 格网的重力部分使用最佳横向重力参数 ... }

表征物理层级的框架按照功能呈层次化定义。3D-横向%、2D-横向以及材料框架分别对应于抗侧力体系配置所作出的决策；而20-水平、支撑边以及支撑分隔框架则对应于重力体系（即楼盖体系）配置所作出的决策。在物理层级中还存在其他框架。

* 这些表示与梁、柱和斜撑等构件相关的信息。

{ 3D-侧向-上-侧向 3D-描述 }

```
{  2D-lateral
            is-alt   3D-lateral
            paxrt-of
            uses
            direction
            2D-description }

{  material
            is-alt  2D-lateral
            mat-description
            dead-load-est  125
            story-dim-est  10.0}

{  2D-horizontal
            is-alt  gravity
            hor-description }

{  support-edges
            is-alt  2D-horizontal
            sup-edges }

{  support-div
            is-alt   support-edges
            subdivide-direction ... }
```

The frames are used to organize the representation of the alternative design solutions. An example of preliminary design solutions generated by HI-RISE is shown in Figure 4-1. The boxes in the Figure repesent frames, and the lines represent relationships between frames. An *is-alt* relationship implies that the connected frames are alternative solutions, a *part-of* relationship implies that all the connected frames are part of one solution, and the *uses* relationship is defined so that information may be inherited horizontally in the tree of solutions.

4.2. FLODER

FLODER [4] is a knowledge based expert system that designs alternative floor systems and framings for a given architectural floor plan. The architectural floor plan can be specified as a set of rectangles representing the building perimeter, hallways, and service shafts. FLODER generates a column grid and proportions the floor system and framing system for gravity loads. FLODER addresses the preliminary design of floor systenns when the column tocations are not fixed and can be specified for maximum efficiency of the structural system.

FLODER requires three different kinds of input from the user: architectural data, structural data, and evaluation data. The architectural data includes geomtetric data and functional data. The user is first required to specify the floor plan in 'rterms of its geometry, i.e. the size and location of rectangles, and function, i.e. hallways and service shafts. The user is then prompted for the type of structural material to

{二维侧向 s - a lt 三维侧向 部分-使用
方向 二维描述}

{ 材料为替代方案 2D-横向 材料描述 恒载估算 125 层高估算 10.0} { 2D-水平为替代方案 重力水平描述 } { 支撑边为替代方案 2D-水平 支撑边 } { 支撑划分替代方案 支撑边 划分方向 ...

}

框架用于组织对备选设计方案的表达。图4-1展示了由HI-RISE生成的初步设计方案示例。图中的方框代表框架，连线表示框架之间的关系。is-alt关系表明相连的框架为互斥的备选方案；part-of关系表明所有相连的框架均构成同一方案的一部分；而uses关系则被定义为允许信息在方案树中沿水平方向进行继承。

4.2. FLODER

FLODER [4] 是一个基于知识的专家系统，用于为给定的建筑平面图设计多种备选楼板体系及框架结构。建筑平面图可由一组矩形表示，这些矩形分别代表建筑外围、走廊和设备竖井。FLODER 会生成柱网，并根据重力荷载对楼板体系与框架体系进行比例分配。当柱位尚未确定且可根据结构体系的最优效率进行布置时，FLODER 能够完成楼板体系的初步设计。

FLODER 需要用户输入三类不同的数据：建筑数据、结构数据和评估数据。其中，建筑数据包括几何数据和功能数据。用户首先需要以几何形式（即矩形的尺寸与位置）以及功能形式（即走廊和竖向管井）来指定平面布局。随后，系统会提示用户选择结构材料的类型。

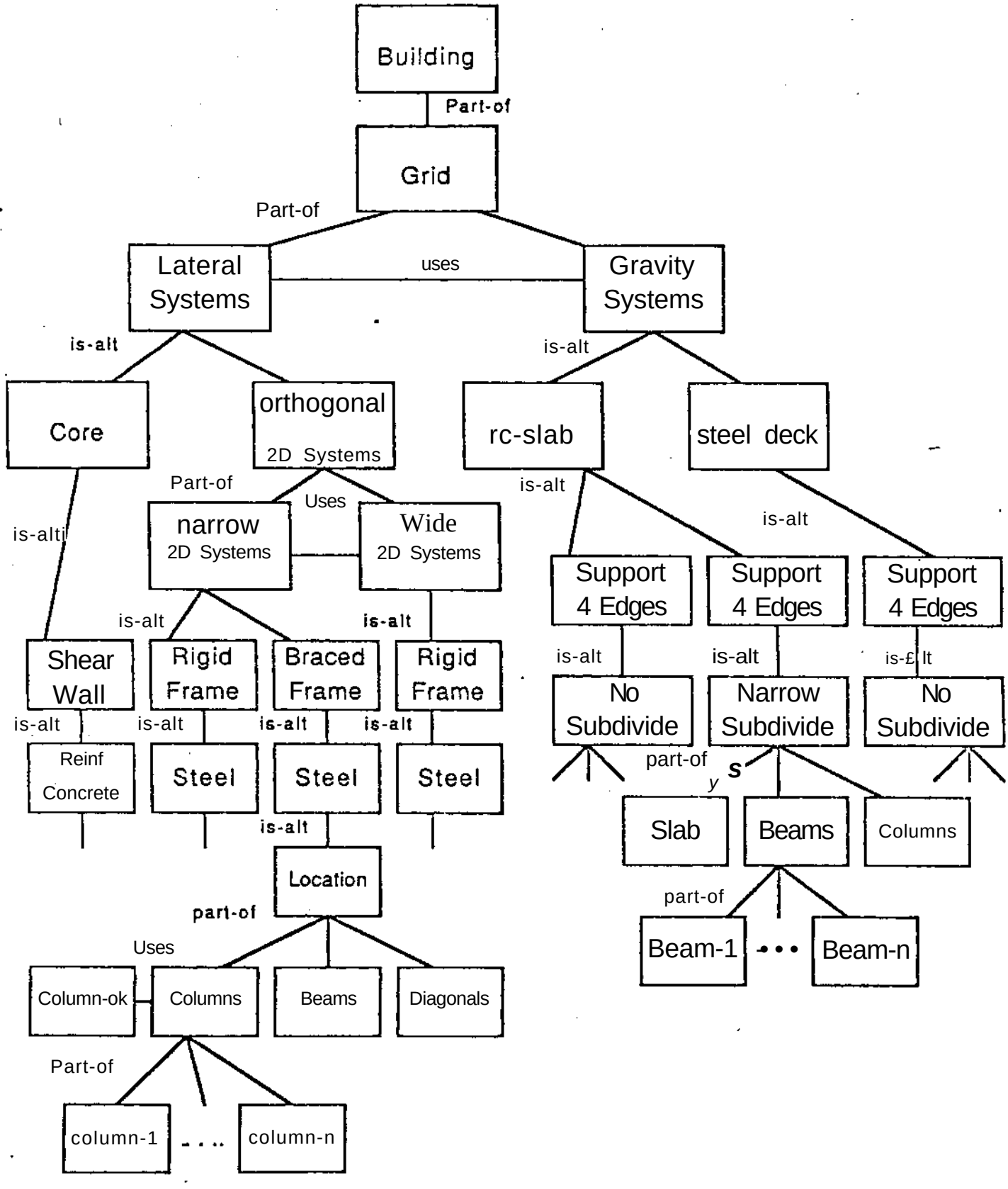


Figure 4-1: Example of Design Solutions in HI-RISE

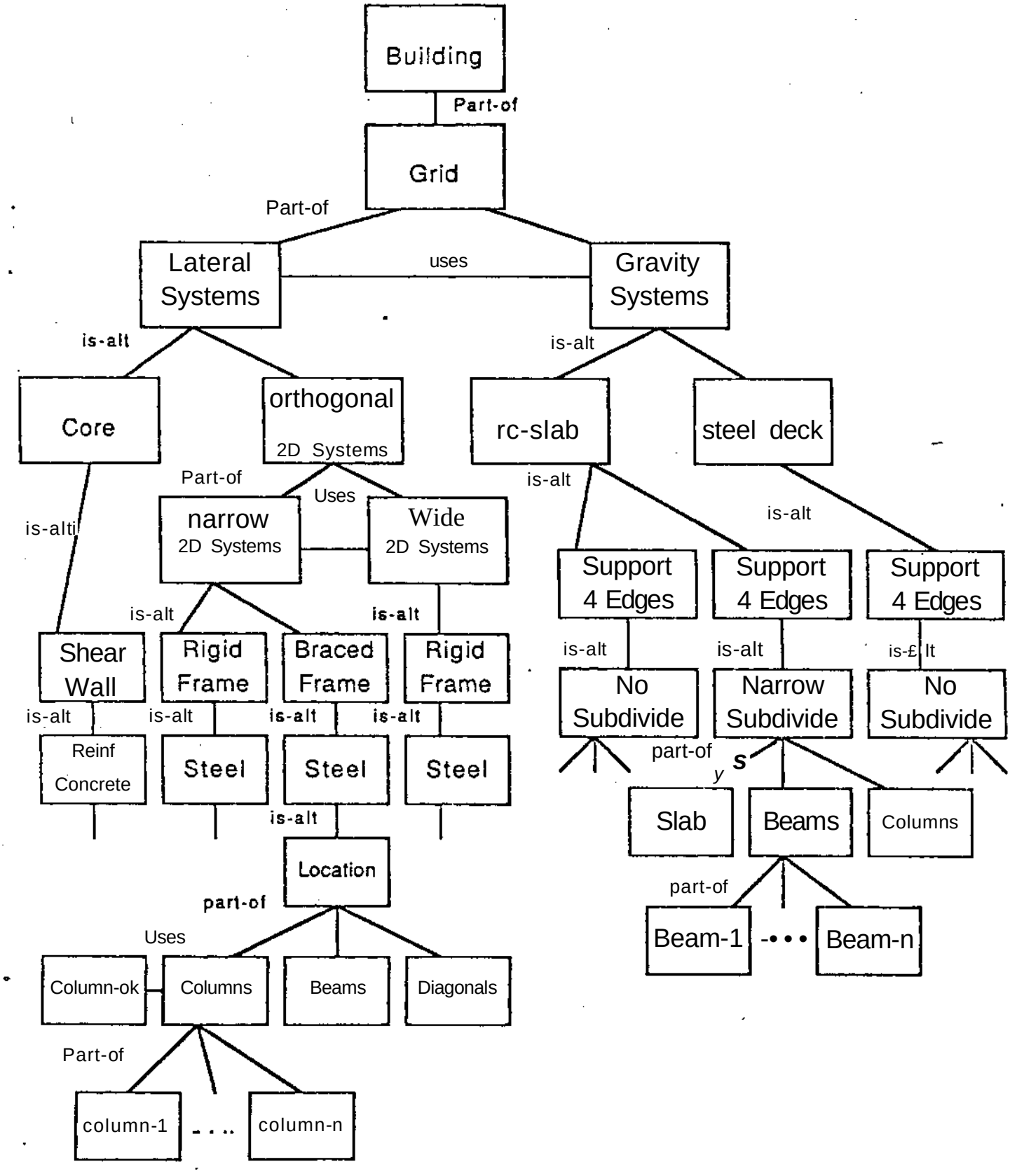


图4-1：HI-RISE中的设计解决方案示例

be used when generating the floor system. The user makes a distinct choice for the 2D horizontal system, e.g. precast panels, and the 1D components, e.g. reinforced concrete. Finally, the *user* is prompted for weighting factors for the properties that are evaluated. The user may generate multiple design solutions by changing the choice of materials for the same architectural data.

The output of FLODER is a graphical representation of the alternative solutions. The graphical representation is provided incrementally; the plan perimeter is illustrated upon input, followed by the column locations, girders (if any), and beams (if any) as they are generated. The first generated floor system is displayed on the whole screen. Subsequent alternatives are displayed on one fourth of the screen and the first alternative is redisplayed.

An example of the graphical presentation of alternative framing plans is shown in Figure 4-2. Each quadrant of the figure illustrates one alternative. The distinction between alternatives is defined by the user when selecting the type of materials to be used. The upper left framing plan was generated using knowledge of reinforced concrete beams, columns and slabs. The upper right framing plan was generated using knowledge of prestressed concrete beams, columns and slabs. The lower left was generated using knowledge of steel beams, columns and steel deck floor systems. The lower right was generated using knowledge of reinforced columns and flat slabs (implies no beams used for support, only columns).

FLODER uses a generate and test method to synthesize a floor plan. The generate stage involves placing the columns, girders, and beams on the floor plan. The test stage involves analyzing portions of the floor system and evaluating the system for constraint satisfaction. These tasks are further described below.

The framing generation proceeds from the placement of the one dimensional elements to the three dimensional configuration. This configuration consists of columns and beams defining a three dimensional frame supporting the two dimensional surface elements. The generation depends on the structural and architectural constraints, since the distance between sequential columns should not exceed the limit for the material type, and no structural elements are allowed in the restricted areas defined by the architectural constraints.

In the analysis subtask, the structural elements to be analyzed are selected as those requiring the biggest and smallest amount of material to meet structural requirements. The analysis module consists of two parts. The first is performed for the comparison of required depths or deflections between elements in the same alternative, while the second part is performed for a comparison of the total amount of material

在生成楼板系统时将采用该方法。用户需分别为二维水平结构（如预制板）和一维构件（如钢筋混凝土）作出明确的选择。最后，系统会提示用户为待评估的各项性能指标设置权重因子。对于同一建筑数据，用户可通过调整材料选择来生成多种设计方案。

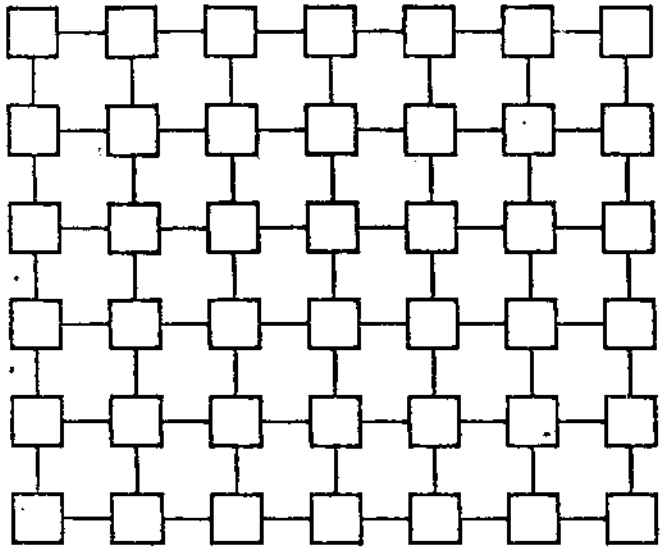
FLODER 的输出为备选方案的图形化表示。该图形化表示以渐进方式呈现：在输入初始信息后首先显示平面轮廓，随后依次展示柱位、主梁（如有）以及次梁（如有）的布置情况。首个生成的楼板体系会占据整个屏幕；后续的备选方案则仅显示于屏幕的四分之一区域，同时原方案将被重新显示。

图4-2展示了一个用于呈现不同结构布置方案的图形示例。图中的每个象限分别对应一种方案。各方案之间的区别由用户在选择所用材料类型时予以界定。左上角的结构布置方案是基于钢筋混凝土梁、柱及板的相关知识生成的；右上角的方案则基于预应力混凝土梁、柱及板的相关知识生成；左下角的方案采用钢梁、钢柱及钢桁架楼盖系统；而右下角的方案则基于钢筋混凝土柱与无梁楼盖（即仅采用柱作为支撑，不设梁）的相关知识生成。

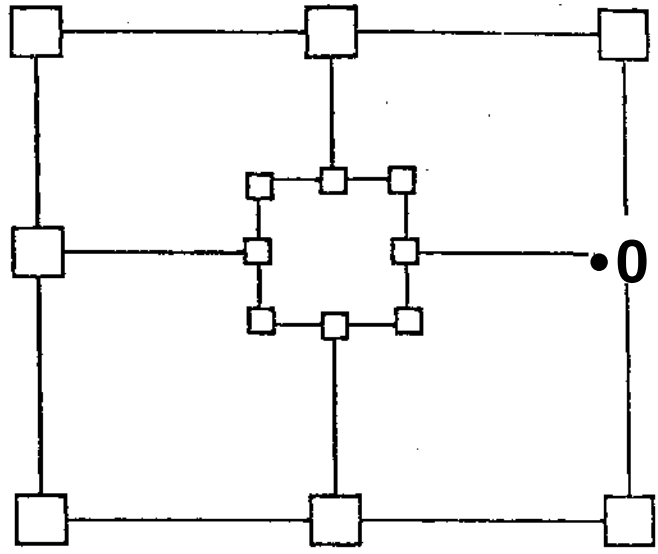
FLODER 采用生成与检验相结合的方法来构建平面布局。在生成阶段，将柱、主梁和次梁布置于平面图中；在检验阶段，则对楼盖系统的各部分进行分析，并评估该系统是否满足约束条件。这些具体步骤将在下文进一步阐述。

框架的生成过程始于一维构件的布置，最终形成三维构型。该构型由柱和梁构成，共同定义了一个支撑二维面状构件的三维框架。其生成过程受结构与建筑约束的双重影响：相邻柱之间的间距不得超过相应材料类型的限值，且在由建筑约束所界定的禁区内不得设置任何结构构件。

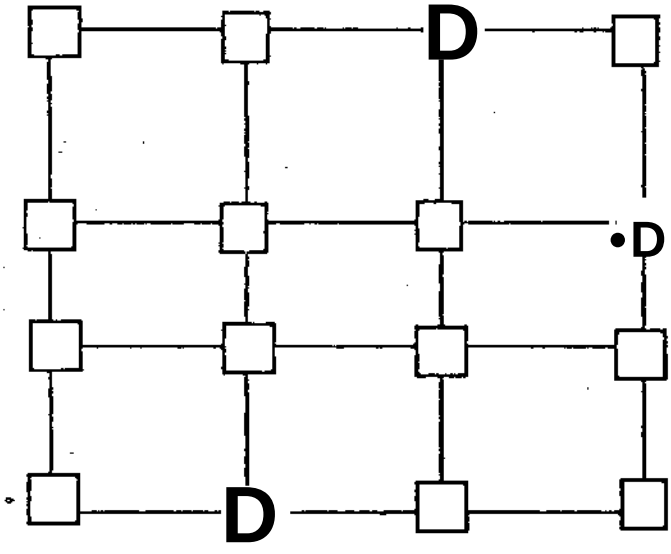
在分析子任务中，待分析的结构构件被选定为那些在满足结构要求时所需材料量最大和最小的构件。分析模块由两部分组成：第一部分用于在同一方案内比较各构件所需的截面高度或挠度；第二部分则用于比较总材料用量。



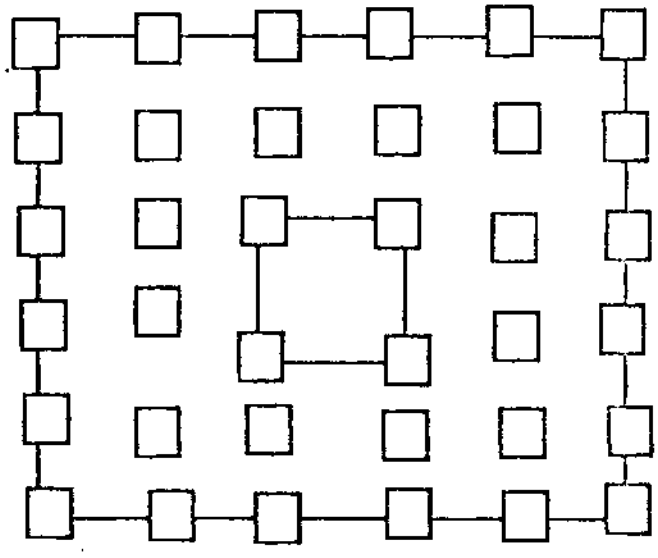
1D-MATERIAL=REINF0RCED CONCRETE
2D-MATERIAL=REINFORCED CONCRETE



1 D-MATERIAL=PRESTRESSED CONCRETE
2D-MATERIAL=PRESTRESSED CONCRETE

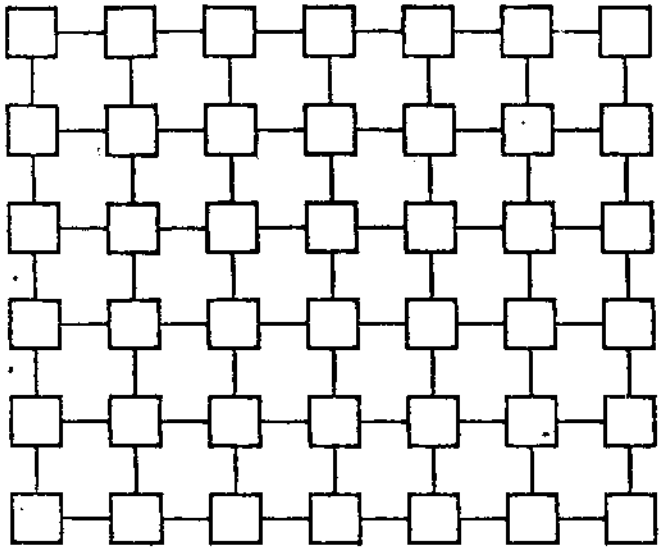


1D-MATERIAL=STEEL
2D-MATERIAL=STEEL-DECK

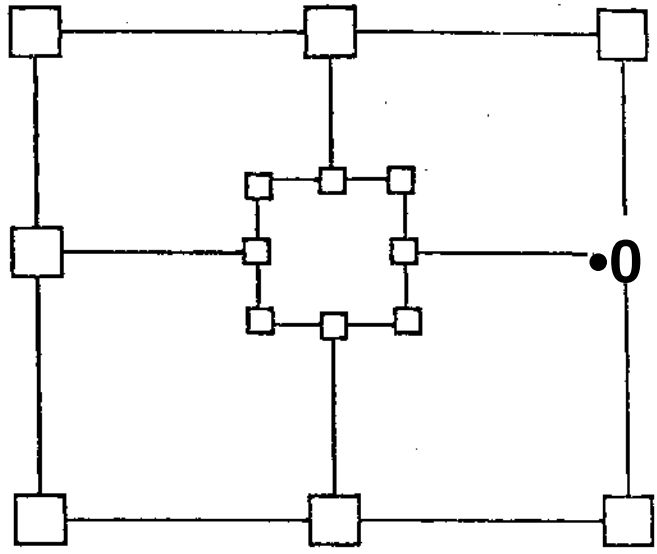


1 D-MATERIAL=REINFORCED CONCRETE
2D-MATERIAL=REINFORCED CONCRETE FLAT SLAB

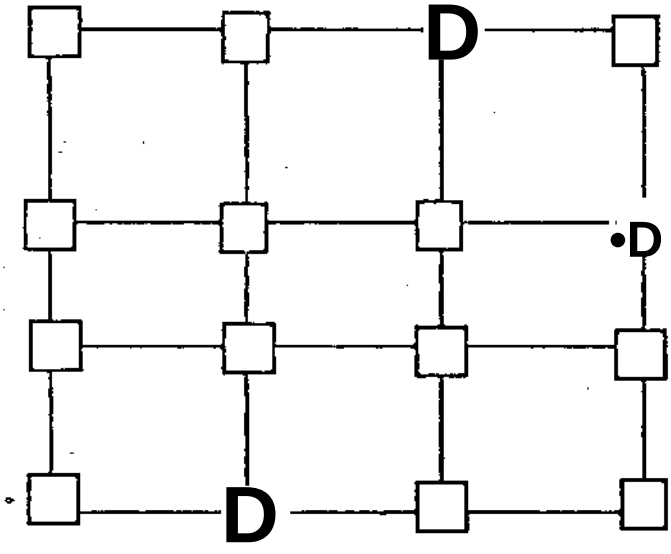
Figure 4-2: Presentation of Framing Plans in FLODER



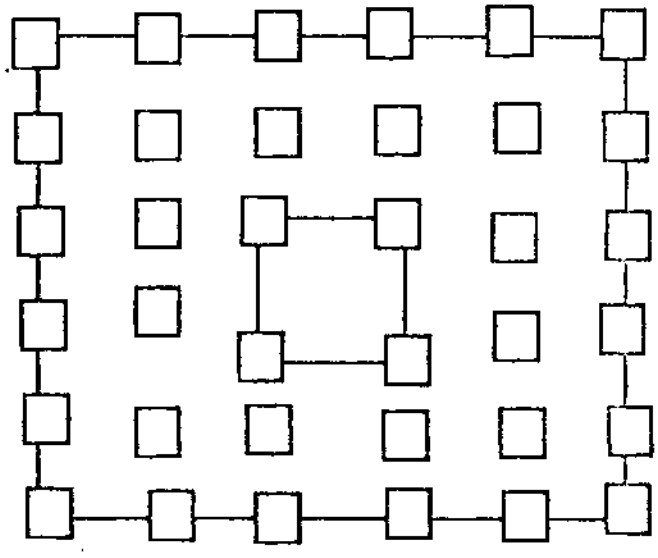
1D-MATERIAL=REINF0RCED CONCRETE
2D-MATERIAL=REINFORCED CONCRETE



1D-MATERIAL=PRESTRESSED CONCRETE
2D-MATERIAL=PRESTRESSED CONCRETE



1D-MATERIAL=STEEL
2D-MATERIAL=STEEL-DECK



1 D-MATERIAL=REINFORCED CONCRETE
2D-MATERIAL=REINFORCED CONCRETE FLAT SLAB

图4-2：FLODER中的框架规划展示

between different alternatives.

The evaluation subtask also consists of two parts; the evaluation of different structural elements within the same framing, and evaluation of alternative framings for the given architectural plan. The first evaluation may result in revisions to a single floor framing, while the second provides information on the relative merits of alternative floor plans.

The knowledge base of FLODER is made up from a combination of OPS5 production rules and Lisp functions. The control and the heuristics are represented by OPS5 rules. Purely algorithmic sections, such as analysis of slabs, are programmed in Lisp which is much easier and faster for manipulating numerical *6c\.?* than OPS5. The user interface module is also coded in Lisp.

4.3. LOCATOR

LOCATOR [8] is an expert system to serve as an assistant in locating lateral load resisting systems on a three dimensional building grid. LOCATOR generates feasible alternatives for the placement of a lateral system configuration specified by the user. The alternatives are tested by checking the satisfaction of heuristic constraints.

The design of the lateral load resisting system is considered in two parts: configuration and layout. Configuration is the process of generating structural systems that can be described in all aspects except their placement on a building grid. In the context of LOCATOR, three choices are made during configuration:

- whether to design a 3D, 2D orthogonal, or combination of a 3D and 2D orthogonal system,
- the structural subsystems to be used, e.g. rigid frames, and
- the structural material to be used.

Layout of a lateral system usually follows its configuration and is concerned with the topology of the design. Once a system is configured, the design must satisfy the spatial or architectural constraints imposed on it. In the context of building design, a building grid is used to represent potential column locations, and layout involves placing the lateral system on this grid.

Upon entering LOCATOR, all configuration decisions are made and specified to the expert system. Locator generates and tests alternative placement schemes for the given configuration and presents them to the'user. The generate and test method is applied to levels of placement decisions: 3D layout, 2D layout in the narrow direction, and 2D layout in the wide direction. At each level, an alternative is selected and tested for feasibility. The alternatives for 3D layout are predetermined according to the type of 3D structural system, for example, the placement of a tube system is on the perimeter of the building. The

在不同备选方案之间。

评估子任务同样包含两个部分：一是对同一框架内不同结构构件的评估，二是针对给定建筑方案的备选框架进行评估。前者可能导致对单层楼板框架的修改，而后者则提供关于不同楼板方案相对优劣的信息。

FLODER的知识库由OPS5产生式规则与Lisp函数相结合构成。控制策略和启发式方法由OPS5规则表示，而诸如板坯分析等纯算法性模块则采用Lisp编程实现，因为在处理数值运算方面，Lisp比OPS5更为简便且高效。用户界面模块同样以Lisp语言编写。

4.3. 定位器

LOCATOR[8] 是一种专家系统，旨在作为辅助工具，用于在三维建筑网格中确定抗侧力体系的位置。LOCATOR 能够为用户指定的抗侧力体系布置方案生成可行的备选方案，并通过检验启发式约束的满足情况来对这些备选方案进行评估。

抗侧力体系的设计可分为两个部分：构型与布置。构型是指在不考虑其在建筑网格中具体位置的情况下，生成能够从各方面加以描述的结构体系的过程。在LOCATOR的框架下，构型阶段需做出三项选择：

- 是设计三维系统、二维正交系统，还是采用三维与二维正交系统的组合?
- 拟采用的结构子系统，例如刚架等，并
- 所选用的结构材料。

侧向体系的布置通常与其构型相一致，并涉及设计的拓扑结构。一旦确定了体系的构型，设计必须满足对其施加的空间或建筑约束。在建筑设计的语境中，建筑网格用于表示潜在的柱位，而布置则是在该网格上规划侧向体系的定位。

进入 LOCATOR 系统后，所有配置决策均被制定并指定给专家系统。Locator 会针对给定的配置生成并测试多种备选布置方案，并将其呈现给用户。该“生成与测试”方法应用于多个层次的布置决策：三维布局、沿窄向的二维布局以及沿宽向的二维布局。在每一层次，都会选择一个备选方案并对其可行性进行验证。其中，三维布局的备选方案已根据三维结构体系的类型预先确定，例如，筒体系统的布置通常位于建筑的外围。

-alternatives for 2D layout are synthesized by combining alternatives for *column lines used and bays used,* Figure 4-3 illustrates some possible combinations for 2D layout.

LOCATOR was developed in Knowledge Craft [2], an environment for building expert systems. LOCATOR makes use of CRL, an object oriented representation language. The design knofwedge is represented in schemata (or frames) and rulesets. Schemata are used to form templates for alternative placement schemes and to explicitly represent decision levels. Rulesets, coded in Lisp, are sets of conditional statements that test intermediate solutions for feasibility.

4.4. STRUPLE

STRUPLE [6, 9] is an expert system for structural configuration planning. STRUPLE uses a database to store structural design solutions collected from several structural consulting firms and construction companies. The major capacity of STRUPLE is: given the description of a building, mainly the architectural information and some structural information such as the loads and the construction material that the designer prefers to use, STRUPLE will find relevant structural solutions in the database and use the information in the old solutions to plan the structural configuration of the new building.

Experiences used by STRUPLE are represented in the form of incomplete descriptions of building design solutions, which gives the following information about buildings.

- General information. This includes building number, name, design completion date, location, owner, general use of the building, and unit and total costs.
- Geometric information. This includes gross area, height, width, aspect ratio, overall shape and shape irregularity which is a real number roughly indicating how irregular the shape is.
- Other architectural specifications. This includes a number of stories below and above grade, intended use, bay numbers and sizes in two main orthogonal directions, floor-to-floor height, ceiling-to-ceiling height, number of service shafts and general description of their location in the build plan.
- Load information. This includes dead load, live load, wind load and seismic zone.
- Primary 3D systems. A building may use one than one type of 3D system, for example, one or more cores with a 2D orthogonal system, which may be composed of rigid frames, tied together.
- Lateral 2D systems. Lateral 2D systems are the subsystems of 3D- systems. The information about lateral 2D systems includes the structural type of 2D systems, the construction materials, and directions in which they are placed.
- Floor systems. This includes types of floor systems, which imply the construction material type, types of support systems, for example, columns in the case that floor system is flat slab, and construction materials of the support systems.
- Foundation systems. This includes types and bearing capacities of the foundation systems. Again, types also indicate the construction material used for the foundation systems.

The relations in the database are defined using the same organization. This description only includes

二维布局的备选方案通过组合所用列线和所用跨距的备选方案来生成，图4-3展示了二维布局的一些可能组合。

LOCATOR 是在 Knowledge Craft [2] 中开发的，Knowledge Craft 是一个用于构建专家系统的环境。LOCATOR 使用 CRL，这是一种面向对象的表示语言。设计知识以模式（或框架）和规则集的形式进行表达。模式用于构建备选布局方案的模板，并显式地表征决策层次。规则集以 Lisp 语言编写，是由一系列条件语句组成的集合，用于检验中间解的可行性。

4.4. STRUPLE

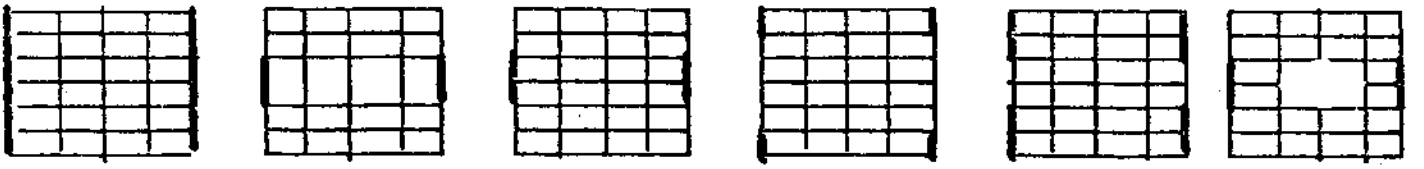
STRUPLE [6, 9] 是一种用于结构方案规划的专家系统。STRUPLE 使用数据库存储从多家结构咨询公司和建筑企业收集的结构设计方案。其主要功能是：给定一栋建筑的描述，主要包括建筑信息以及一些结构信息，如荷载和设计者偏好的建筑材料，STRUPLE 将在数据库中检索相关的结构方案，并利用已有方案中的信息为新建筑的结构布置提供规划建议。

STRUPLE 所采用的经验以建筑设计方案的不完整描述形式呈现，从而提供关于建筑的以下信息。

- 一般信息。这包括建筑编号、名称、设计竣工日期、所在地、业主、建筑的总体用途以及单位造价和总造价。
- 几何信息。这包括总面积、高度、宽度、长宽比、整体形状以及形状不规则度，其中形状不规则度为一个实数，用于大致表征形状的不规则程度。
- 其他建筑规格。这包括地下及地上楼层数、建筑用途、沿两个主要正交方向的开间数量与尺寸、层高、净高、服务竖井的数量及其在建筑平面图中的大致位置描述。
- 荷载信息。这包括恒载、活载、风荷载以及地震设防烈度。
- 主要的三维结构体系。一座建筑可能采用一种以上的三维结构体系，例如，一个或多个核心区域采用二维正交体系，该体系可由刚性框架构成，并相互连接。
- 横向二维系统。横向二维系统是三维系统的子系统。关于横向二维系统的信息包括二维系统的结构类型、所用建筑材料以及其布置方向。
- 楼板系统。这包括楼板系统的类型，即所采用的建筑材料类型、支撑体系的类型（例如，在楼板为无梁楼盖的情况下，支撑体系为柱），以及支撑体系的建筑材料。
- 地基系统。这包括地基系统的类型及其承载能力。同样，类型也反映了地基系统所采用的建筑材料。

数据库中的关系采用相同的组织方式定义。本描述仅包含

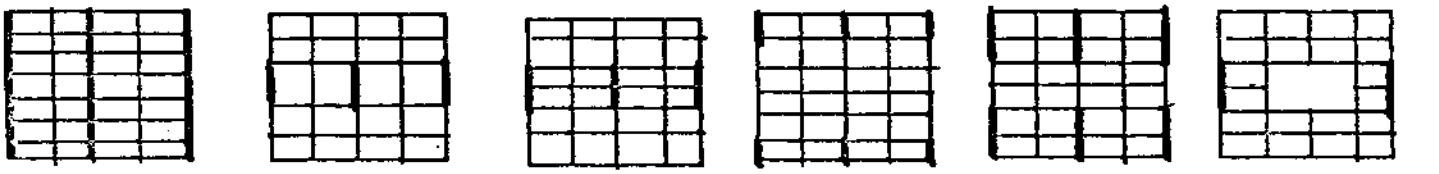
COLUMN UNES USED: EDGES



COMPLETE INNER-1 INNER-2 OUTER-1 OUTER-2 INNER-SHAFT

BAYS USED

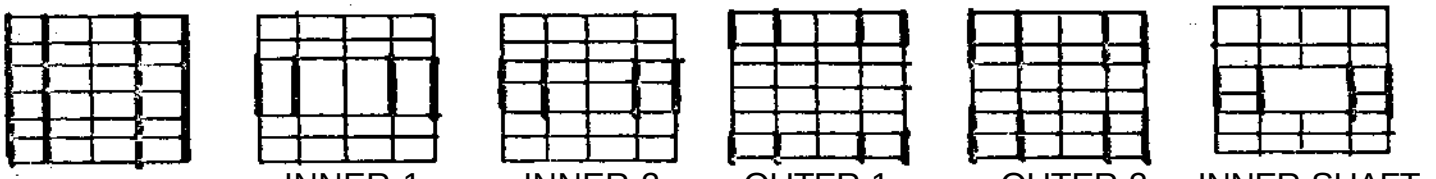
COLUMN LINES USED: EDGES +1



COMPLETE INNER-1 INNER-2 OUTER-1 OUTER-2 INNER-SHAFT

BAYS USED

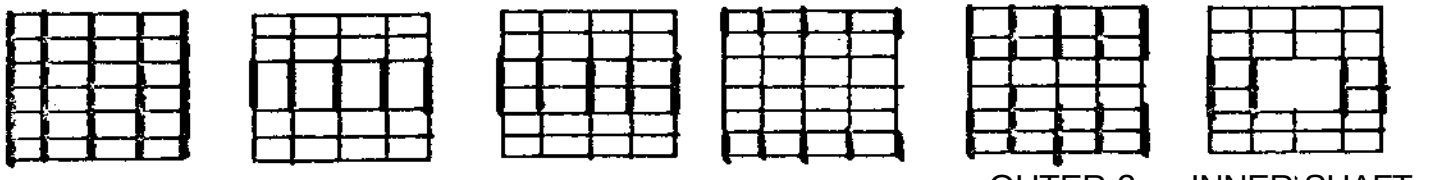
COLUMN UNES USED: EDGES+2



COMPLETE INNER-1 INNER-2 OUTER-1 OUTER-2 INNER-SHAFT

BAYS USED

COLUMN LINES USED: EDGES+ALL



COMPLETE INNER-1 INNER-2 OUTER-1 OUTER-2 INNER-SHAFT

BAYS USED

Figure 4-3: 2D Layout Alternatives in LOCATOR

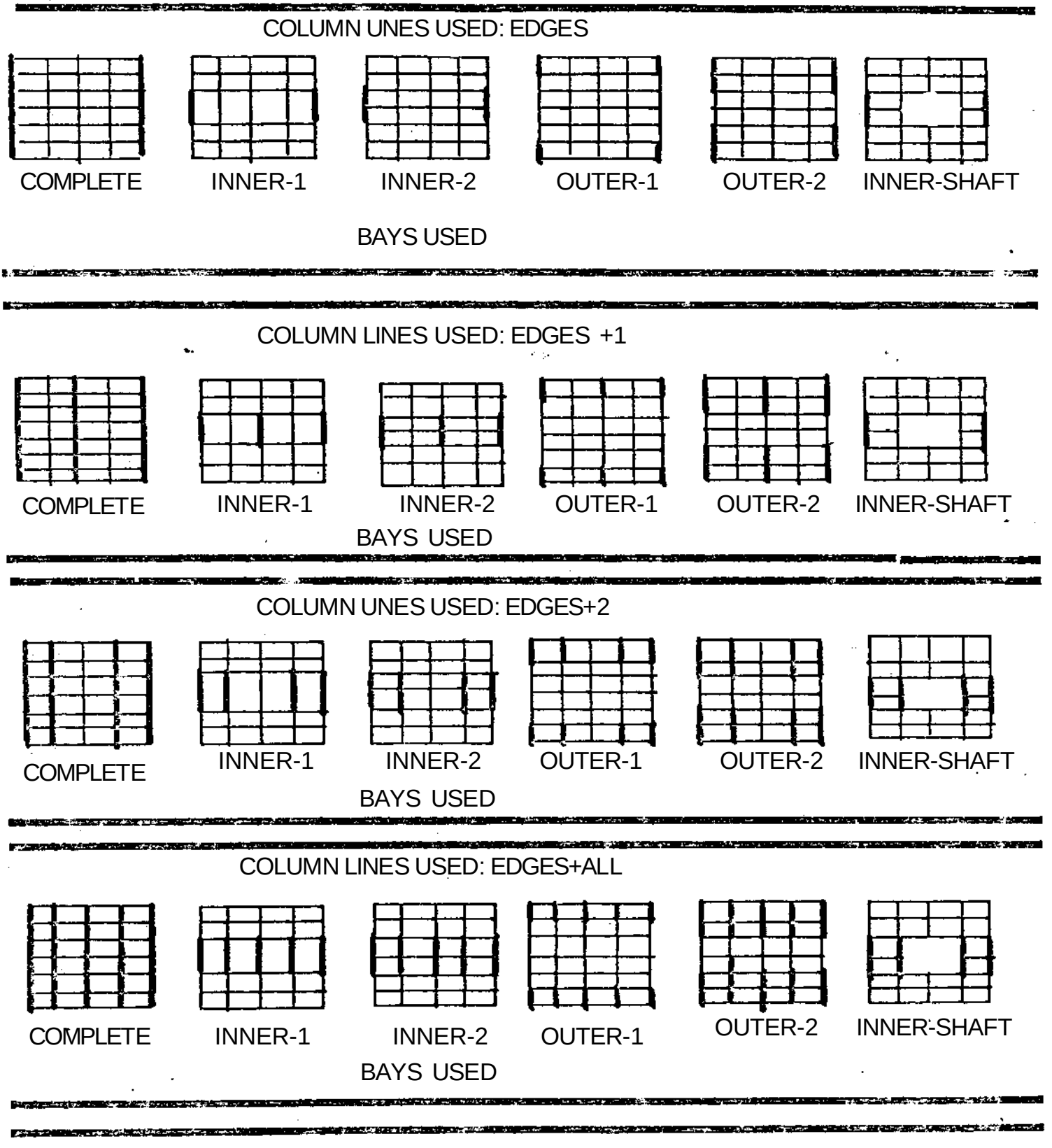


图4-3：LOCATOR中的二维布局方案

the general characteristics of buildings. Some information that is important, as well as more detailed information, is missing because either it is difficult to represent or is not important to this study.

The first task of STRUPLE is to find similar experiences to the current design problem. The input is a partial description of the architectural plan and some structural information of a building to be designed. STRUPLE generates a list of similar buildings retrieved from the database. The evaluations of the buildings and their rankings based on their similarities to the input building are also included.

The buildings which are considered to be similar to the given building are called *matching buildings,* or *matches.* To find matches requires a similarity metric to determine which buildings in the database are similar to the input building. The similarity metric is described by a set of *criteria,* also called *matching criteria,* which define what significant common aspects the matching buildings and the input building should share. A matching criterion is a requirement of similarity imposed on a *feature* of a matching building. For example, the number of stories, the intended use, the design wind toad, the construction material, etc. all are features of a building and are potential criteria. Obviously, not all features of a building are equally important in terms of their impacts on the structural system. Thus, criteria are divided into three classes: required criteria, desired criteria and no-match criteria, where required criteria must be satisfied by a matching building, the satisfaction of desired criteria is desirable but not necessary and no-match criteria are not considered. *Required, desired* and *no-match* are qualifiers that represent the *status* of a criterion.

While searching the database for similar buildings, required criteria are used as qualifiers. In order to locate a matching building explicitly, we need to know the limits within which the building can be considered to be a match. Thus a *range* must be set for each criterion. For a required criterion, the range is defined by the limits of allowable difference of the feature values of the two buildings compared. To measure how well the matching buildings resemble the given building, each match is evaluated and ranked. A *single evaluation* is a measure of the difference between the values of one feature of a match and the given building. The *final evaluation* is a comprehensive measure combining all individual evaluations for a match.

Once matching buildings have been found which represent the experience that may be relevant to the new design problem, we have to determine what knowledge is to be transformed and used to solve the new problem. It is a fact that even though two buildings could have many common aspects, because of their different special design requirements, the design solutions may not be the same. However, it is also a fact that a certain group of design vocabulary frequently appears in certain types of buildings. Although we can not directly use the design solution of an existing building for a new building, we can use the

建筑物的一般特征。一些重要信息以及更为详细的信息由于难以表达或与本研究无关而缺失。

STRUPLE 的首要任务是为当前的设计问题寻找相似的工程实例。其输入包括待设计建筑的部分建筑方案描述以及若干结构信息。STRUPLE 会从数据库中检索并生成一份相似建筑的列表，同时给出各建筑与输入建筑的相似度评价及其排序结果。

被视为与给定建筑相似的建筑被称为匹配建筑，或简称匹配。为了找到匹配建筑，需要采用一种相似性度量方法，以确定数据库中的哪些建筑是

- 与输入建筑相似。相似性度量由一组标准来描述，这些标准也称为匹配准则，用于界定待匹配建筑与输入建筑应具备的哪些重要共性。匹配准则是对匹配建筑某一特征施加的相似性要求。例如，层数、用途、设计风荷载、建筑材料等均为建筑的特征，也是潜在的匹配准则。显然，就其对结构体系的影响而言，并非建筑的所有特征都具有同等的重要性。因此，匹配准则被划分为三类：必要准则、期望准则和不予考虑准则；其中，必要准则必须由匹配建筑满足，期望准则的满足虽为理想但非强制性要求，而不予考虑准则则不予纳入考量。必要、期望和不予考虑是用于表征匹配准则状态的限定词。

在数据库中检索相似建筑时，将所需条件用作筛选依据。为了明确地找到匹配的建筑，我们需要确定在何种范围内可将其视为匹配对象，因此必须为每一项条件设定一个取值范围。对于一项必要条件，其取值范围由两栋建筑相应特征值之间允许差异的上下限来界定。为衡量匹配建筑与给定建筑的相似程度，每项匹配都会被评估并进行排序。单次评估即为某项特征在匹配建筑与给定建筑之间差异的量化指标；最终评估则是在综合各项单项评估结果的基础上，对某一匹配建筑作出的整体评价。

一旦找到了与新设计问题可能相关的、具有相似特征的既有建筑，我们就必须确定哪些知识需要被转化并用于解决该新问题。事实上，即使两座建筑在许多方面存在共性，由于其各自独特的设计需求不同，设计方案也可能并不相同。然而，这也

。某一设计语汇在特定类型建筑中频繁出现这一事实。尽管我们不能直接将既有建筑的设计方案照搬到新建筑中，但我们可以利用

design vocabulary of the old design to construct the new solution. Thus we choose the design vocabulary as the knowledge to be extracted from the old design solution and be transformed.

The database may have a vast amount of design solutions and many of them may be found to be similar to the new building. In order to use the information efficiently, the frequency of each element of design vocabulary used in the old designs is calculated. According to-their frequencies of appearance and evaluations based on their similarity to the new building, each type of element is assigned a priority of consideration. A very high priority indicates that this type of element is frequently used in the buildings of the type of the new one and should be considered first.

STRUPLE has a set of structural elements that are stored in its knowledge base representing the complete design vocabulary that the synthesis process knows about. The structural elements are organized into different abstraction levels. For each level of abstraction, STRUPLE determines and orders the design vocabulary to be considered for the new building, thus identifying a subset of the complete vocabulary of structural elements which is most promising. This subset of design vocabulary will be used during the synthesis process, in which each structural element will be examined to determine whether it is an eligible or efficient alternative.

STRUPLE is implemented in OPS83 [3], a language developed for expert system applications at Carnegie Mellon University. OPS83 "combines the rule-based programming paradigm of the earlier versions of OPS with the procedural programming paradigm of conventional programming languages" [1]. In OPS33, in addition to working memory elements, other data types and data structures present in Pascal are available. The basic program units are production rules and procedures. Users can determine and implement their own inference mechanism and conflict resolution strategies by writing necessary procedures with the help of some working memory retrieval functions buitt in OPS83. These features of OPS83 makes it more flexible and easier to build the user interface. The database used in STRUPLE is implemented by Ingres, a relational database management system. The communication between the database and the expert system is set up through a C procedure call which is quite straight forward.

5. Summary and Conclusions

Four expert systems for preliminary structural design were presented. HI-RISE was the first system developed and serves as the basis for current research. HI-RISE identifies a fomnalization for preliminary structural design as a synthesize, analyze, and evaluate paradigm. HI-RISE illustrates that although computer aids can be developed for preliminary design, much work needs to be done before such aids

利用旧设计方案的设计词汇来构建新的解决方案。因此，我们选择将设计词汇作为从旧设计方案中提取并进行转化的知识。

数据库中可能包含海量的设计方案，其中许多方案与新建筑具有较高的相似性。为了高效利用这些信息，我们计算了旧设计方案中各设计要素的出现频率。根据其出现频率以及与新建筑的相似度评价，为每类设计要素赋予相应的优先级。优先级越高，表明该类要素在与新建筑同类型的建筑中使用频率较高，应优先予以考虑。

STRUPLE在其知识库中存储了一组结构元素，这些元素构成了合成过程所知的完整设计词汇。这些结构元素被组织为不同的抽象层次。对于每个抽象层次，STRUPLE都会确定并排序待用于新建筑的设计词汇，从而识别出最具潜力的结构元素子集。在合成过程中，将使用该设计词汇子集，并对其中的每个结构元素进行评估，以判断其是否为合格且高效的备选方案。

STRUPLE 是在 OPS83 [3] 中实现的，OPS83 是一种由卡内基梅隆大学为专家系统应用开发的语言。OPS83“将早期 OPS 版本的基于规则的编程范式与传统编程语言的面向过程编程范式相结合”[1]。在 OPS83 中，除了工作内存元素之外，Pascal 中存在的其他数据类型和数据结构也可直接使用。其基本程序单元是产生式规则和过程。用户可以通过编写必要的过程，并借助 OPS83 内置的一些工作内存检索函数，自行确定并实现推理机制与冲突消解策略。这些特性使 OPS83 更加灵活，也更便于构建用户界面。STRUPLE 所使用的数据库由关系型数据库管理系统 Ingres 实现，数据库与专家系统之间的通信通过 C 语言的过程调用建立，过程十分直接。

5. 总结与结论

介绍了四个用于初步结构设计的专家系统。HI-RISE是最早开发的系统，也是当前研究的基础。HI-RISE将初步结构设计的形式化描述归纳为“综合—分析—评估”范式。HI-RISE表明，尽管可以为初步设计开发计算机辅助工具，但在这些工具真正实用化之前，仍有大量工作需要完成。

can be practical. FLODER is an expert, system that addresses the decisions made in generatirg a structural grid for a building. LOCATOR proposes a methodology for generating alternative placement schemes for a lateral toad resisting system on a given grid. STRUPLE serves as a preprocessor to the synthesis part of the preliminary design by using past experience to identify the structural subsystems and materials relevant to the current design.

In conclusion, the use of expert system techniques holds much promise for structural designers. Through the development of expert systems for preliminary design, designer's are asked to identify their own design process. This identification serves to improve the designer's and the engineering community's understanding of the design process. The use of expert systems can improve the quality of the preliminary design process .by bringing more experience to bear on the current problem than that of a single designer.

可以具有实用性。FLODER 是一个专家系统，用于处理在生成建筑结构网格时所作出的决策。LOCATOR 提出了一种方法，可在给定网格上生成侧向荷载抵抗体系的多种布置方案。STRUPLE 作为初步设计中综合阶段的预处理器，利用以往的设计经验识别与当前设计相关的结构子系统和材料。

总之，专家系统技术在结构设计领域具有广阔的应用前景。通过开发用于初步设计的专家系统，设计人员被要求对其自身的设计过程进行梳理与归纳；这一过程有助于增进设计人员及整个工程界对设计流程的理解。借助专家系统的应用，可以通过汇集比单一设计人员更为丰富的经验，从而提升初步设计的质量。

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